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Education

Indian Institute of Technology (ISM) Dhanbad

Oct 2021 – May 2025

Bachelor of Technology in Mining Engineering

Dhanbad, Jharkhand

- **Relevant Coursework:** Artificial Intelligence, Machine Learning (Supervised & Unsupervised), Deep Learning, Natural Language Processing, Probability & Statistics, Data Analysis, Data Visualization, Python Programming.

Experience

Outlier AI

Nov 2024 – Nov 2025

Machine Learning Engineer – Freelance

Remote

- Worked as a **Machine Learning Engineer (AI Model Trainer)** on large-scale **LLM evaluation and improvement projects**, with emphasis on **model reasoning quality, correctness, and robustness**.
- Designed and executed **structured evaluation pipelines** to assess LLM outputs, including **response ranking, chain-of-thought analysis**, and systematic **failure-mode identification**.
- Reviewed and debugged **LLM-generated Python code**, identifying **logical, syntactic, and performance issues**; applied **machine learning and analytical techniques** to improve code correctness and reliability.
- Built **Python-based automation scripts** using **pandas, numpy, and SQL** to analyze **error patterns, label inconsistencies, and model behavior at scale**.
- Generated **high-quality training signals and feedback** used in **fine-tuning workflows**, directly contributing to improvements in model **accuracy, consistency, and reasoning performance**.
- Collaborated asynchronously with **distributed teams**, delivering precise **technical feedback** aligned with real-world **machine learning engineering scenarios**.

EXTION

Feb 2025 – May 2025

Machine Learning Intern

Remote – Uttar Pradesh, India

- Designed and implemented a production-ready **NLP pipeline in Python** using **NLTK, VADER, Hugging Face Transformers, and PyTorch** to classify **financial news headlines** into sentiment classes, achieving **92%+ accuracy**.
- Performed extensive **text preprocessing and feature engineering**, including **tokenization, stopword removal, and lemmatization**, to improve model generalization and robustness across noisy real-world data.
- Analyzed sentiment outputs at scale and built **visual analytics dashboards** using **Matplotlib and Seaborn** to uncover **market sentiment trends** and volatility signals.
- Optimized **GPU-accelerated inference pipelines** using **CUDA and cuDNN**, reducing **end-to-end latency by 40%** and enabling near **real-time processing of 10k+ headlines**.

Projects

Smart GenAI PDF Chatbot | RAG, FastAPI, LangChain, FAISS, Docker, GPT-3.5

🐙 GitHub

- Extracted multi-page PDF text using **pdfplumber**, applied **spaCy** sentence tokenization for clean chunking, and generated dense vector embeddings via **all-MiniLM-L6-v2** (SentenceTransformers) indexed into a **FAISS IndexFlatL2** store.
- Built a hybrid query engine with tunable L2 distance threshold — high-confidence matches serve answers directly from PDF; low-confidence queries route through **LangChain PromptTemplate + LLMChain** to **GPT-3.5-turbo** fallback, achieving 100% routing accuracy.
- Exposed the pipeline via a production **FastAPI** backend with **/upload, /ask, /stats, and /health** endpoints, per-request **JSON monitoring (query_logs.json)**, latency tracking (**~23 ms avg**), and **A/B prompt testing framework**.
- Containerized the full stack with **Docker** for one-command reproducible deployment (**docker build + run**), ensuring environment-consistent execution across local and cloud targets.

Bhagavad Gita NLP Chatbot | Semantic Search, Multilingual NLP, FAISS, Gradio

🐙 GitHub

- Built a multilingual semantic-search chatbot supporting **Hindi, English, and Sanskrit** using **Sentence-BERT** embeddings.
- Indexed and queried **700+ Bhagavad Gita verses** using **FAISS** for concept-based, context-aware retrieval.
- Deployed on **Hugging Face Spaces** with **Gradio** for seamless real-time interaction without paid APIs.
- Achieved **90%+ answer relevance** on philosophical Q&A benchmarks; earned a **Kaggle Bronze Medal**.

NTT Stock Price Prediction | Machine Learning, LSTM, Time-Series Forecasting

🐙 GitHub

- Built an end-to-end time-series forecasting pipeline with robust preprocessing and feature engineering.
- Trained and evaluated **8+ ML models** (Linear Regression, Random Forest, CatBoost, etc.) achieving up to **95.46% R²**.
- Developed an **LSTM neural network** using **TensorFlow (Keras)** to capture long-term temporal dependencies.
- Forecasted a 30-day stock price trend (**\$178.97 → \$102.04**) with **RMSE of 3.09**.

Technical Skills

Languages: Python, SQL
AI/ML: Supervised & Unsupervised Learning, Deep Learning, NLP, LLMs, Prompt Engineering, RAG, Semantic Search (FAISS, SBERT), Time-Series Forecasting
Frameworks/Libraries: Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, FAISS, Sentence-Transformers, LangChain, FastAPI, spaCy, pdfplumber, Pandas, NumPy, Matplotlib, Seaborn
Data Analytics & BI: Power BI, DAX, Power Query, Data Modeling (Fact–Dimension), Excel (Data Cleaning & Analysis)
Deployment & Tools: Git/GitHub, Docker, GitHub Actions (CI/CD), Streamlit, Gradio, Hugging Face Spaces, AWS (Basic), Azure App Service

Achievements

- **Kaggle:** 3× Bronze Medals in NLP Chatbot, E-Retail Segmentation, and CNN Image Classification.
- **HackerRank:** Earned a **5 rating in Python** by solving multiple coding challenges.
- **HackerRank:** Earned a **5 rating in SQL** by solving advanced database queries and problems.

Leadership / Extracurricular

- Organizer – Khanan’22 Geo-Mining Fest, IIT Dhanbad
- Completed **BCG X Data Science Virtual Experience Program (Forage)**
- Volunteered with **Kartavya NGO** (teaching underprivileged children)
- Cricket enthusiast